How-To Library

Informatica

FAQ: Authentication on HDInsight with Enterprise Security Package

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Abstract

Customers of Microsoft Azure and Informatica can integrate Data Engineering 10.4.x with an HDInsight compute cluster and associated ADLS storage resources. The integration allows users to run mappings and workflows on HDInsight to access data from and write data to ADLS.

This article contains frequently asked questions about managing users and addressing common authentication issues in the HDInsight with ESP environment.

Supported Versions

- Data Engineering Integration 10.4 x

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Overview

You can integrate Data Engineering 10.4 with a Microsoft Azure HDInsight Hadoop cluster with Azure Data Lake Store (ADLS) storage and the Enterprise Security Package (ESP).

HDInsight clusters with ESP are connected to an Azure Active Directory (AAD) domain, which controls users, roles, groups, and permissions for the Azure platform and its clients. Informatica can authenticate AAD domain credentials to enable users to access the cluster and run jobs.

Use the answers and tips in this article to prevent or solve authentication and access issues.

Frequently Asked Questions

This section contains frequently asked questions and frequently encountered issues around issues of authentication and access.

How Do I Set Up Users to Run Mappings?

HDInsight clusters with Enterprise Security Package (ESP) can take advantage of robust authentication by using Azure Active Directory (AAD) users. You can use role-based access control (RBAC) policies or Access Control Lists (ACLs) to control access to storage resources.

AAD is a service that provides single sign-on and multi-factor authentication for users who connect to the Azure cloud platform. AAD uses role-based access control policies and access control lists to enable administrators to manage user authentication and authorization.

Set up users and groups in the Azure Active Directory (AAD) before you create Virtual Machines (VMs), SQL Server databases, and the HDInsight cluster to run jobs in the Azure environment.

When you create an HDInsight cluster with access to ADLS Gen1 or ADLS Gen2 as primary storage, synchronize AAD users to enable them to run jobs on the cluster. Azure synchronizes HDInsight cluster users with AAD if the users are members of groups with permissions to create and run jobs on the cluster.

Verify that the Service Principal Display Name (SPN) used during cluster creation is assigned to the IAM role on the ADLS storage that is associated with the cluster. Mappings might fail if the SPN is not assigned to the IAM role.

How Do I Generate a Keytab File for Users?

A .keytab file contains the shared secret key of an Azure service. Create a keytab file on any one of the cluster nodes for each Informatica user.

A keytab is a file containing pairs of Kerberos principals and encrypted keys which are derived from the Kerberos password. You can use a keytab file to authenticate Informatica users on the cluster without entering a password.

To generate the keytab, use the **ktutil** utility.

Before you begin, get the Kerberos principal user name from the cluster administrator.

1. Log in to any cluster VM.
2. From the command line, type `ktutil` to launch the utility.
3. Type the following command:

   ```
   addent -password -p <user name> -k 1 -e RC4-HMAC
   ```

   where <user name> is the Kerberos principal user. For example:

   ```
   addent -password -p myname -k 1 -e RC4-HMAC
   ```
4. When prompted, enter the password for the Kerberos principal user.
5. Type the following command to create a keytab:

   ```
   wkt /tmp/keytabs/<user name>.keytab
   ```

   where `/tmp/keytabs/` is the path where you want to store keytabs. For example:

   ```
   wkt /tmp/keytabs/myname.keytab
   ```
6. Type `q` to quit the ktutil utility.

Example

```
sshuser@hn0-hivesc:/tmp/keytabs$ ktutil
ktutil: addent -password -p alicetest -k 1 -e RC4-HMAC
Password for alicetest@SECUREHADOOPRC.ONMICROSOFT.COM:
ktutil:    wkt /tmp/keytabs/alicetest.keytab
ktutil: q
```

Why Can't I Find AD Users for the HDInsight Cluster?

The Azure Active Directory (AAD) provides the pool of available users. To run jobs on the HDInsight cluster, AAD must be synchronized with the cluster.

Users from Azure AD will be synchronized to the cluster only if they are part of HDInsight cluster creation group “Cluster user groups.”

Note: Ranger permissions can be assigned only to cluster user groups, even though Ranger shows all groups.

Can Informatica Use AD Connect Users?

Yes, users from AD connect are supported in Informatica.

User Impersonation Isn't Working. What's Wrong?

To enable the Informatica user to run jobs on the HDInsight cluster, add the proxyuser properties to the HDFS service

Use the Ambari HDInsight cluster interface to add the following properties to the HDFS service, where <proxy user> is the Service Principal Name associated with the keytab user:

hadoop.proxyuser.<proxy user>.hosts

Set to a single host name or IP address, or set to a comma-separated list. For less security, use the wildcard " * " to allow impersonation from any host.

hadoop.proxyuser.<proxy user>.groups

Set to group names of impersonation users separated by commas. For less security, use the wildcard " * " to allow impersonation by any group.

hadoop.proxyuser.<proxy user>.users

Set to a single user account or set to a comma-separated list. For less security, use the wildcard " * " to allow impersonation by any user.

Restart the cluster credential service after adding these properties.

To run Sqoop mappings on the Spark engine, add the Hadoop impersonation user as a Linux user on the machine that hosts the Data Integration Service.

See the list of types of users and the instructions to grant access to ADLS resources to Informatica users in the *Data Integration Guide.*

Troubleshooting

If user impersonation isn't working, verify that you restarted the cluster credential service after you added proxyuser properties.

You can also examine the cluster credential service log, located in `/var/log/hdinsight-credentialservice/`.

What Permissions Are Required to Run a Big Data Mapping?

Grant read and execute permission for the staging user/impersonation user to access ADLS Gen1 or ADLS Gen2 storage.

You can use role-based access control (RBAC) or access control lists (ACLs) to grant permissions. Use RBAC to grant extensive permissions to access ADLS. Use ACLs to grant permissions to user groups at a more granular level.

See the following topics in the *Data Engineering Integration Guide:*

- Grant Access Permissions to ADLS Gen1 Storage

- Grant Access Permissions to ADLS Gen2 Storage

Which Users Require Tokens?

The cluster uses an OAUTH token for authentication for the impersonation user, the Hadoop staging user, and any other user who runs jobs on the cluster from the Data Integration Service.

An access token is an object that represents the authorization of a specific application to access specific resources. Informatica uses tokens to authenticate access to the HDInsight cluster and ADLS storage resources.

The following users need tokens to execute a mapping:

- Keytab user
- Impersonation user
- Hadoop staging user
- Blaze Engine user
- OS Profile user
- Logged-in user, if you use the logged-in user as the Impersonation user in an OS Profile.

For definitions of these users, see the Data Integration Guide.

Note: The OAUTH token is only required if a mapping accesses ADLS Gen1 resources.

To generate the OAUTH token for authentication for the Informatica users, log in to Ambari Web UI with the Azure Active Directory user credentials. See the Azure documentation.

Mappings Fail with a TokenNotFoundException

The Data Integration Service uses the ADLS/ABFS driver to retrieve the OAUTH access token from the Azure credential service. If the token is not present, the mapping fails with a TokenNotFoundException.

Verify that you have generated the token for each Informatica user. See "Which Users Require Tokens?" on page 5

Mappings are Not Being Submitted to the Cluster

If the mapping session log shows that mappings are not being submitted to the cluster, verify that you have completed the following setup tasks:

Identify Hosts

Edit the `/etc/hosts` file on the machine that hosts the Data Integration Service to include the IP address, DNS name, and DNS short name for each node of the cluster. Use `headnodehost` to identify the host as the cluster headnode host.

Copy Security Files

Copy the krb5.conf file to the following locations on the machine that hosts the Data Integration Service:

- `<Informatica installation directory>/java/jre/lib/security`
- `<Informatica installation directory>/services/shared/security`

Copy the keytab file to the following directory: `<Informatica installation directory>/isp/config/keys`.

Configure Data Integration Service Properties

Verify that the following Data Integration Service properties are correctly configured:
Hadoop Kerberos Service Principal Name

Service Principal Name (SPN) created for the user.

Hadoop Kerberos Keytab

The file path to the Kerberos keytab file on the machine that runs the Data Integration Service.

Hadoop Staging User

An HDFS user who performs operations on the cluster staging directory. Ensure that the user has the required permissions on the cluster staging directories such as Auto install etc..

How Can I Debug Mapping Failures?

Most of the failures with ESP are reproducible outside Informatica. To simplify access related problems, you can use a Java program to debug failures.

Edit the Java Program

Download the Java program from the following link: TesHdiAdlsFS.zip

Uncompress and edit the program to provide values for the following variables:

- Keytab file location on the cluster
- Impersonation user
- Location of *-site.xml files on the cluster
- HDFS location on ADLS or WASB storage

Uncomment the delegation token section and provide user names.

Run the Java Program

The following commands provide an example of how to run the program:

```
export CLASSPATH=<Informatica installation directory>/services/shared/hadoop/
HDInsight_<distribution_version>/lib/*:.
javac -cp $CLASSPATH -Xlint:unchecked TesHdiAdlsFS.java
java -cp $CLASSPATH TesHdiAdlsFS
```

Must the Cluster Configuration be Modified to Run Mappings?

No. After you import the cluster configuration (CCO) from the HDInsight cluster with ESP and ADLS storage to the Data Integration Service, no editing of cluster configuration properties is required.

Is Reading from and Writing to HDFS Supported?

Yes, Informatica supports reading from and writing to the HDFS associated with the cluster.

To enable access to HDFS, edit the HDFS connection and supply the value of fs.defaultFS. You can find the value for fs.defaultFS in the core-site.xml configuration set of the cluster configuration.

Note: If you create connections when you import the cluster configuration, the NameNode URI property is populated by default, and it is updated each time you refresh the cluster configuration. If you manually set this property or override the value, the refresh operation does not update this property.

See the Data Integration Guide for more information about HDFS Connection Properties.

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